

SOPHIA DA COSTA FERNANDES

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PROFESSIONAL OBJECTIVE

To work on building efficient systems in a software engineering environment, enabling the development of real solutions and contributing to the delivery of clean and scalable code.

PROFESSIONAL SUMMARY

Computer Engineering student focused on Backend development and software architecture. Corporate experience at Stellantis in automotive embedded systems, ECU validation and cloud data integration. Track record of self-directed hands-on projects with Java, Spring Boot, IoT, APIs and mobile development, including work awarded Best Practical Project of the 1st semester at PUC Minas. Fluent in English.

PROFESSIONAL EXPERIENCE

Stellantis - Engineering Internship *June 2025 – Present (12 months)*

- Development of an embedded solution for collecting data from test benches and vehicles, using Raspberry Pi + ESP32 + MCP2515, currently in use by the C&I team in Electrical Engineering.
- Validation of automotive systems with a focus on Infotainment (Radio and Cluster), including failure analysis and debugging of embedded systems to ensure the integrity of subsystem communication over the CAN network.
- Integration of telemetry flows with cloud infrastructure in the context of the connected vehicles team, ensuring real-time data synchronization between the vehicle and external platforms.

PROJECTS

Cod Invest - Financial Asset Management Platform | 2026 *(in development)*

- Full-stack application for tracking and analyzing investment funds with API integration (BrapiDev) for consuming real financial data. Focused on end-user experience and clean code best practices, with Java, JavaScript, Bootstrap and Spring Boot.
- Repository: github.com/sophiafernandesc/cod-invest-project

Diagnosticar - Vehicle Diagnostics IoT System | 2025

- Full-stack IoT system for real-time vehicle diagnostics: data pipeline with embedded acquisition over CAN (Arduino Nano + MCP2515), ESP32 bridge, MQTT broker for cloud infrastructure and presentation layer in an Android application developed in React Native.
- Repository: github.com/sophiafernandesc/esp32-canbus-obd2-project

Voltra - Fuel Management and Vehicle Tracking Application | 2025

- Android application developed in Flutter for fuel management and automotive tracking, reading consumption parameters via an ELM327 scanner integrated with an ESP32 with a tracking module, with data persistence in MySQL.
- Repository: github.com/sophiafernandesc/voltra-project

Spindle - Quadruped Robot | 2024

- Quadruped robot with 8 servomotors for articulation and locomotion, developed in C++, with a mobile control application developed in APP Inventor. Awarded Best Practical Project of the 1st semester (2024/1) in Computer Engineering at PUC Minas.

- Repository: github.com/sophiafernandesc/spindle-project

EDUCATION

Computer Engineering | PUC Minas - *In progress. Expected completion: December 2028.*

Extracurricular Activity: Quantum Computing Study Group (2nd semester of 2024), focused on exploring the theoretical foundations and applications of the field.

SKILLS AND COMPETENCIES

Backend: Java, Spring Boot, REST APIs, OOP, Clean Code, MySQL

Frontend / Mobile: JavaScript, React Native, Flutter, Bootstrap

Embedded / IoT: C/C++, Arduino, ESP32, CAN Bus, MQTT

Cloud / DevOps: MQTT Broker, cloud infrastructure integration

Tools: Git, GitHub, Linux (Red Hat)

CERTIFICATIONS

Red Hat Linux Course: Fundamentals of Linux-based operating systems.

Java Course - Udemy: Study focused on programming logic, Object-Oriented Programming and software structuring.

Automotive Networks - PUC Minas: Technical study of embedded communication protocols: CAN, LIN and Automotive Ethernet networks, with a focus on systems integration and vehicle connectivity with cloud services.

LANGUAGES

English - Fluent: reading, writing and conversation